CMAI 480

Computer Vision

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Outline

- Geometric primitives and transformations
  - Review of Linear Algebra
  - Projection geometry
    - Homogeneous Representation of 2D/3D entities or features
    - 2D/3D Image Transformations
  - Camera geometry
- Human vision system
- Human vision for computer vision
- Cameras and image formation

PROJECTION GEOMETRY

HOMOGENEOUS REPRESENTATION OF WORLD 2D ENTITIES

Homogeneous Representation

- It is a representation of physical world entities where only ratios matters.

- Let's recall the equation of line *l* using its slope *m* and a point (x_0, y_0) that lies on it.

$$y - y_0 = m(x - x_0)$$

$$\rightarrow mx - y + (-mx_0 + y_0) = 0$$

$$\boxed{\rightarrow ax + by + c = 0}, \text{ where } a = m, b = -1, c = (-mx_0 + y_0)$$

Homogeneous Representation

- Note that the equation of a line $ax + by + c = 0$ can be written in a Matrix vector multiplication as follows:

- $$[a \quad b \quad c] \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0, \qquad \underbrace{[a \quad b \quad c]}_{line\ Par.} \overbrace{\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}}^{Point\ Par.} = 0, \text{ or } \boldsymbol{l^T \vec{X} = 0}$$

Points, Lines and Conics
Homogeneous Coordinates

- Homogeneous Coordinates based representations for geometrical entities lead to compact (and easy to program) formulas for many useful operations on the entities.
- Let $\mathbb{R}^2$ be a 2D space of reals.
- A line in $\mathbb{R}^2$ is a set of points $(x, y) \in \mathbb{R}^2$ that obey $ax + by + c = 0$ for some values for the parameters a, b, and c.
- Let's call $l = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ as our 3D parameter vector
- Let's represent the 2D physical point (x, y) by a 3D vector $X = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$

Points, Lines and Conics
Homogeneous Coordinates

- Obviously, the algebraic equation of a line, $ax + by + c = 0$, can be expressed as: $l^T X = 0$ or $X^T l = 0$
- The 3D vector $X = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \in \mathbb{R}^3$ is the homogeneous coordinate representation of a physical 2D point $(x, y) \in \mathbb{R}^2$.
- Exploring the 3D representational space $\mathbb{R}^3$, we note that for any multiple $k \in \mathbb{R},\ k \neq 0,\ k. \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$ is the same physical point in $\mathbb{R}^2$ as that corresponding to $\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$. [That is certainly the case with regard to the use of points in the algebraic form $ax + by + c = 0$.]

Points, Lines and Conics
Homogeneous Coordinates

- So in the $\mathbb{R}^3$ representational space, all the points $k.\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$, with $k \neq 0$,

  form an equivalence class, in the sense that they represent the same physical point $(x, y) \in \mathbb{R}^2$. That is, each radial line passing through the origin, but not including the origin, in $\mathbb{R}^3$represents a unique physical point in $\mathbb{R}^2$.
  - [Q: How is the origin of the physical $\mathbb{R}^2$space represented in the $\mathbb{R}^3$ HC space?]
    - It should be $(0,0,1)^T$, but as we will see later this is reserved for a very special entity that is yet to be described.
  - HC: From now on, let's use HC as an abbreviation for " homogeneous coordinates".

Points, Lines and Conics
Homogeneous Coordinates: Recap

- If we represent a point in 2D physical space $\mathbb{R}^2$ by $(X, Y)^T$.
  - Its homogeneous coordinates in $\mathbb{R}^3 - (0,0,0)^T$ will be denoted by the vector $(x_1, x_2, x_3)^T$ with $X = x_1/x_3$, $Y = x_2/x_3$.
    - The notation $-(0,0,0)^T$ indicates that the vector $(0,0,0)^T$, which doesn't correspond to any point/line, is excluded.

Points, Lines and Conics
Homogeneous Coordinates

- Let's now talk about HC representation for lines in physical $\mathbb{R}^2$.
- Referring again to the algebraic form $ax + by + c = 0$, it turns out that the $\mathbb{R}^3$ space, in which the parameter vectors $l = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ are situated, can serve as the HC representational space for the physical lines in $\mathbb{R}^2$.
- Paralleling what we saw for the HC representation of physical points in $\mathbb{R}^2$, all vectors $k.\begin{pmatrix} a \\ b \\ c \end{pmatrix}$, $k \neq 0$, in $\mathbb{R}^3$ represent the same physical line in $\mathbb{R}^2$.

Points, Lines and Conics
Homogeneous Coordinates

- Again paralleling the HC representation of physical points, every radial line in $\mathbb{R}^3$ (that is used for the parameter vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$), but without the point at its origin, represents a unique line in physical $\mathbb{R}^2$.
- Here are a couple of immediate payoffs of our HC representation for points and lines in physical $\mathbb{R}^2$:

Points, Lines and Conics

Homogeneous Coordinates: Payoffs of our HC Representation

1. Given any two lines $l_1 = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$ and $l_2 = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$, the point of intersection of the two lines is given by $X = l_1 \times l_2$ where "$\times$" is the vector cross product.
2. Given any two points $X_1 = \begin{pmatrix} x_1 \\ y_1 \\ 1 \end{pmatrix}$ and $X_2 = \begin{pmatrix} x_2 \\ y_2 \\ 1 \end{pmatrix}$, the line that passes through the two points is given by $l = X_1 \times X_2$.

What's l for $x-axis$? $l = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{origin} \times \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{(1,0)\,pt} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

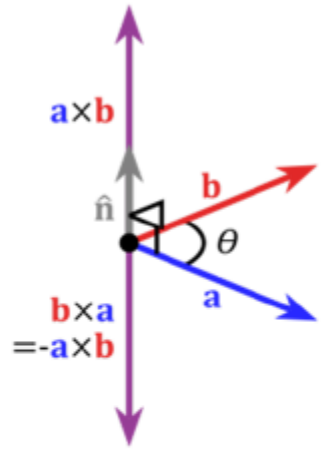
Points, Lines and Conics

Homogeneous Coordinates: Payoffs of our HC Representation

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Recall: Dot Product Vs. Cross Product



$$\mathrm{a}\cdot\mathrm{b} = |\mathrm{a}||\mathrm{b}|\cos\theta,$$

$$\mathrm{a}\times\mathrm{b} = \det\begin{bmatrix} \mathrm{i} & \mathrm{j} & \mathrm{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = \mathrm{i}(a_2 b_3 - a_3 b_2) - \mathrm{j}(a_1 b_3 - a_3 b_1) + \mathrm{k}(a_1 b_2 - a_2 b_1) = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ -(a_1 b_3 - a_3 b_1) \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

Parallel Lines

- Let's now talk about parallel lines: Since the slope of a physical line in $\mathbb{R}^2$ is given by the ratio a/b (or, more precisely, by the pair (a,b)), the two distinct lines $l_1 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $l_2 = \begin{pmatrix} a \\ b \\ c' \end{pmatrix}$ must be parallel.
- You can easily show that the point of intersection of these two lines is $\begin{pmatrix} b \\ -a \\ 0 \end{pmatrix}$. But what does it mean when the third coordinate in the homogeneous representation of a point is zero?

Parallel Lines

- To answer the question in red above, let's consider an arbitrary point $\begin{pmatrix} u \\ v \\ w \end{pmatrix}$ in $\mathbb{R}^3$ representational space for the homogeneous coordinates of the $\mathbb{R}^2$ physical points. Assuming $w \neq 0$, the coordinates of the physical point are given by $(u/w, v/w)$. It is obvious that as $w \to 0$, the x and y coordinates of the physical point more farther and farther away from the origin in $\mathbb{R}^2$ and approach infinity. The important thing to note here is that this approach to infinity is along a specific direction in $\mathbb{R}^2$, the direction being controlled by the values of u and v. When, say, $u = 1$ and $v = 0$, the approach to infinity will be along the $x - axis$ in $\mathbb{R}^2$.

Parallel Lines

- Therefore, the intersection of two parallel lines $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\begin{pmatrix} a \\ b \\ c' \end{pmatrix}$ is at $\begin{pmatrix} -b \\ a \\ 0 \end{pmatrix}$, which is a point at infinity in $\mathbb{R}^2$, along a specific direction controlled by the pair (a,b).
- The physical points in $\mathbb{R}^2$ whose HC representations are of the form $\begin{pmatrix} u \\ v \\ 0 \end{pmatrix}$ are known as **Ideal Points**.

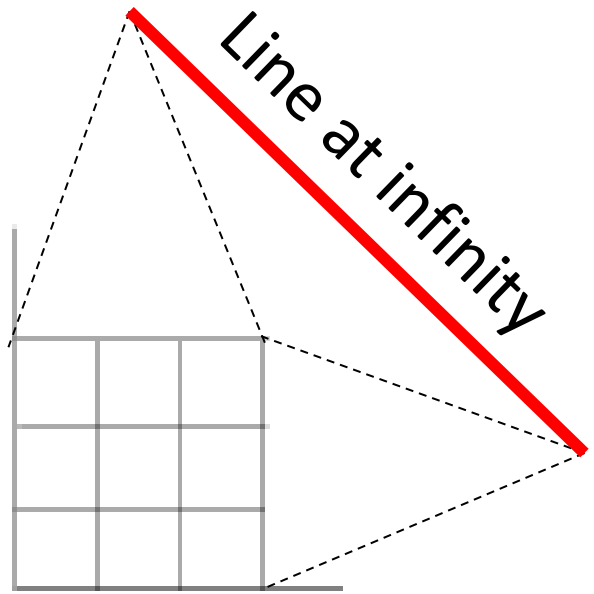
Ideal Points

- **Here us something that could keep you awake at night:** All ideal points from a straight line in $\mathbb{R}^3$.
- To prove the above, consider any two ideal points in $\mathbb{R}^2$: $\begin{pmatrix} u_1 \\ v_1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} u_2 \\ v_2 \\ 0 \end{pmatrix}$. The line that passes through these two points is $l = \begin{pmatrix} u_1 \\ v_1 \\ 0 \end{pmatrix} \times \begin{pmatrix} u_2 \\ v_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ which we get by keeping in mind the notion of equivalence classes in the representation space $\mathbb{R}^3$. Being independent of (u_1, v_1) and (u_2, v_2) parameters, this line remains the same for all pairs of ideal points.

Line at infinity

- The unique line $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is referred to as the Line at Infinity and denoted l_∞.
- The l_∞ line will play an important role in our camera models. The camera image of the l_∞ line is called the **Vanishing Line.**

Line at infinity Vs. Vanishing Line



Real Scene

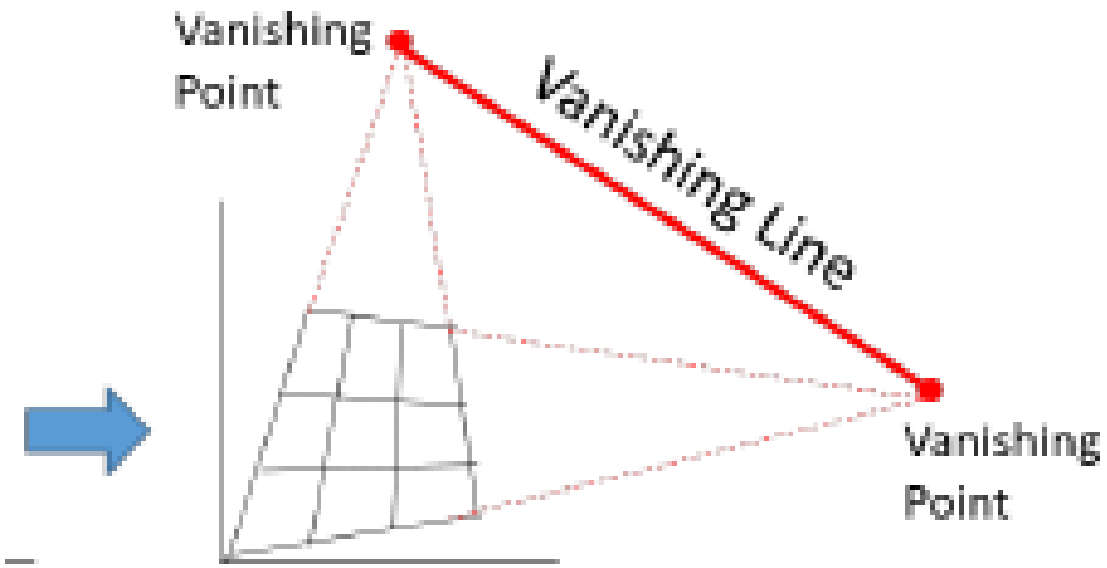


Image of the real Scene
Distortion Effects

Important Observation

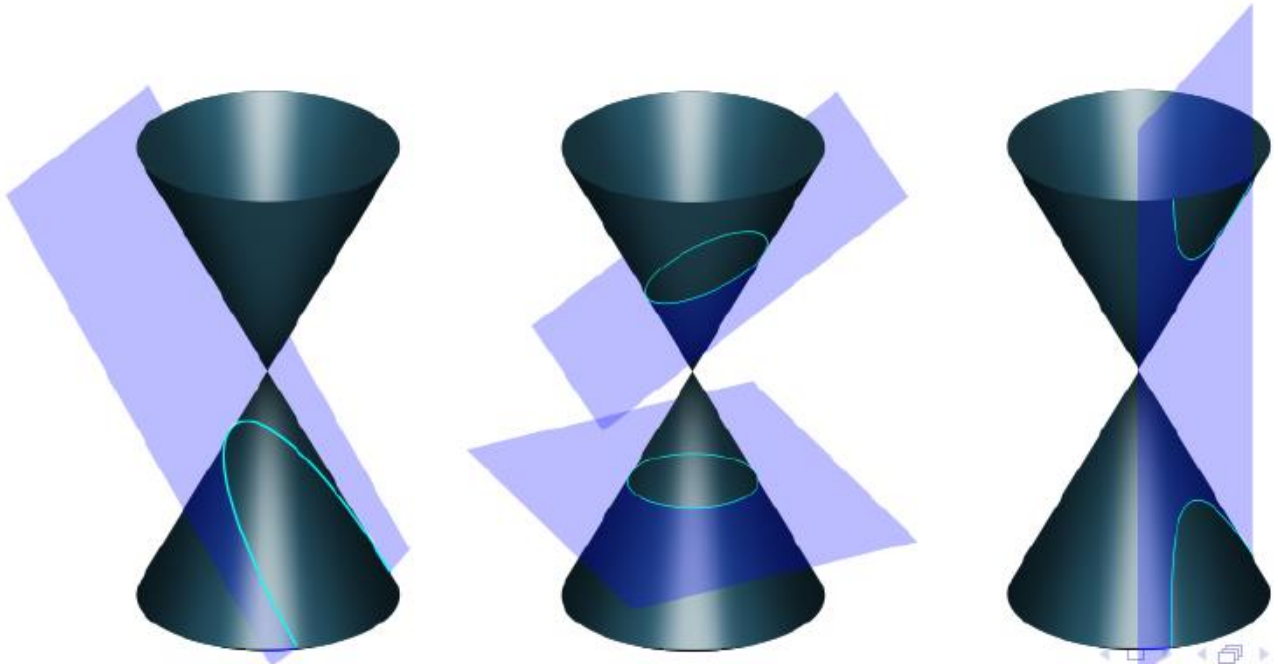
- Did you notice that HC not only give us compact and easy to program formulas, but also that we can represent entities at infinity precisely, that is without approximation and without error.

Points, Lines and Conics
Homogeneous Coordinates

- There is one more 2D geometric entity that will prove important to us in camera modeling and camera calibration work: **Conics**.
- A conic is a curve that is described by a second-degree implicit algebraic form in a plane.
- Euclidean geometry talks about the following three main types of conics: ellipse, hyperbola, and parabola (circle is a special case of ellipse)
  - All of these conics are obtained when you slice a double cone with a plane. A double cone consists of two cones that are vertically aligned and that make a one-point contact at their vertices. When the slicing plane cuts through only one of the cones, you get either a circle, or an ellipse, or a parabola. When the plane slices through both cones, you get a hyperbola.

Points, Lines and Conics
Homogeneous Coordinates

- Types of conic sections:
  1. Parabola
  2. Circle and ellipse
  3. Hyperbola



Points, Lines and Conics
Homogeneous Coordinates

- In the physical plane $\mathbb{R}^2$, the implicit form for a conic is:
  - $ax^2 + bxy + cy^2 + dx + ey + f = 0$
- To express this implicit form in HC, we will first express the point coordinates (x, y) as a 3-vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ with $x = x_1/x_3$ and $y = x_2/x_3$.

  Substituting these expressions for x and y in the implicit form, we get:
  - $ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$

Points, Lines and Conics
Homogeneous Coordinates

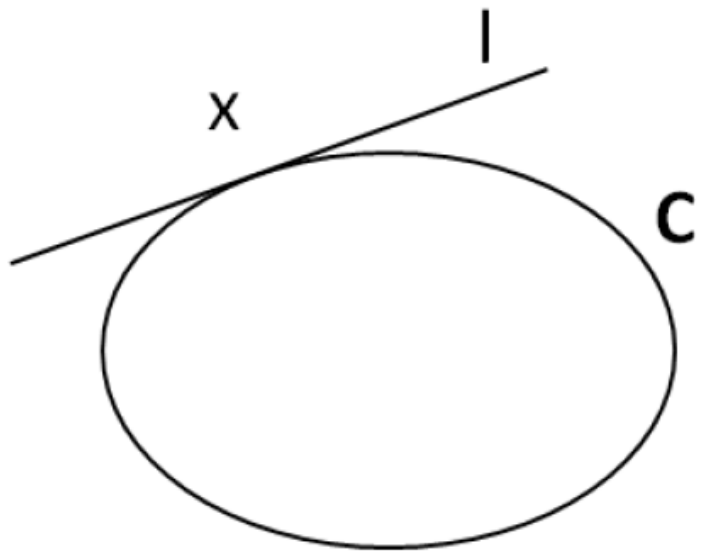
- Rewriting this as a vector-matrix product, we get:
- $\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$, or as
  - $X^T C X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ and $C = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$
    - This is the equation that proves that a given point X is on the conic C.
- C is the HC representation of a conic.

Points, Lines and Conics
Homogeneous Coordinates

- Note that, as with the HC representations for points and lines, only the ratios of the matrix elements are important in the definition of C.
  - In other words, for all $k \neq 0$, $k.C$ represents the same conic as C.

Tangent Lines of a Conic

- The HC representation of a conic gives us compact formulas for the tangent lines to a conic.
- The line l that is the tangent to a conic C at point X (all entities expressed in HC) is given by $\underline{l = CX}$.



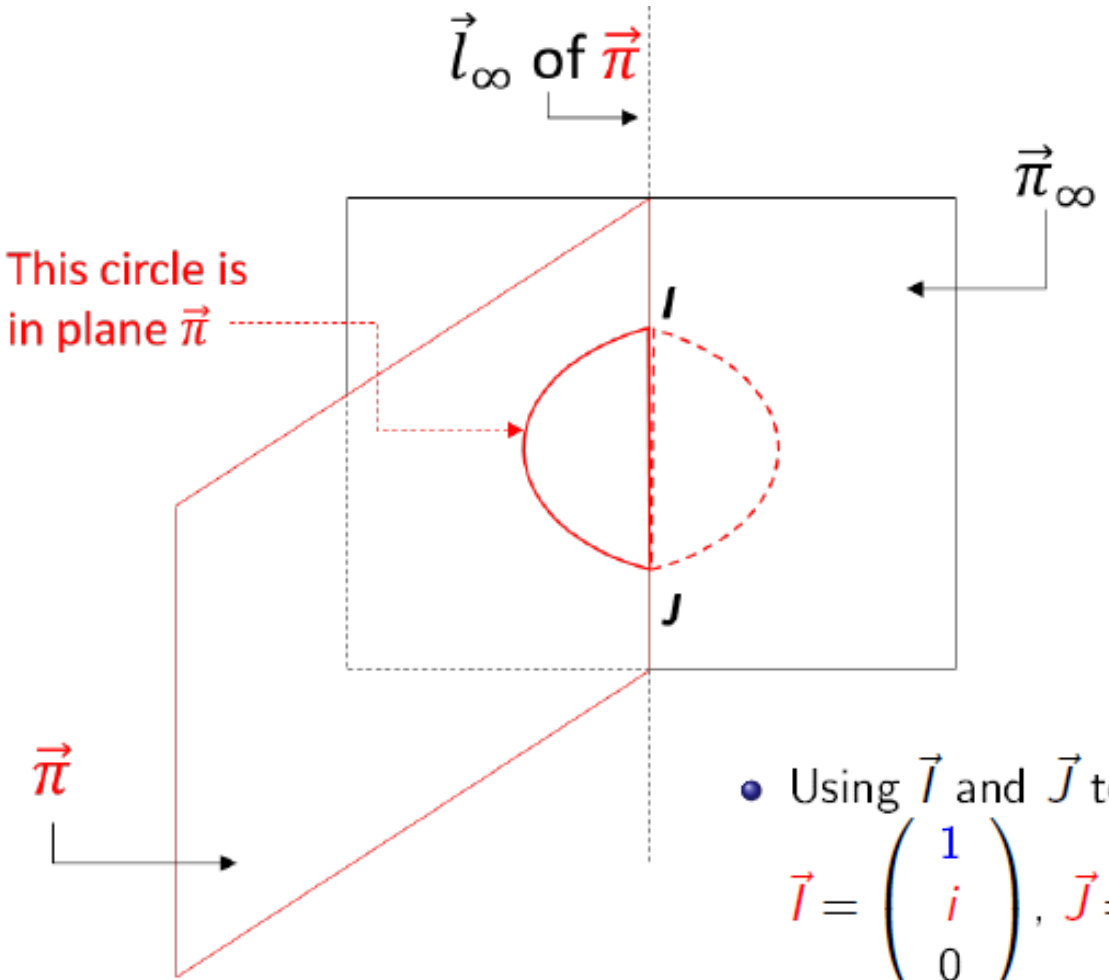
Degenerate Conics

- We will now consider Degenerate Conics.
  - When you slice the double cones with a plane that passes through the axis of the double cones, instead of seeing a hyperbola, you will now see two intersecting straight lines.
    - We say that these two lines constitute a degenerate conic.
  - Let l and m be the HC representations of the two lines.
    - That is, every point on the degenerate conic C is either on line l or on line m. We can show that such a C is given by:
    - $C = lm^T + ml^T \longleftarrow$ Degenerate Conic
- It is trivial to show that such a C satisfies the conic equation $X^T CX = 0$ for all X that are on l, that is for which $l^T X = 0$, and for all X that are on m, that is for which $m^T X = 0$.
  - Note: Each of the two terms in the above addition is a **vector outer product**.
  - Also note that since each outer product has a rank of only 1 (because every column is constant times every other column), **the rank** of C is 2.

Circular Points

- These going to seem like the oddest things, you would have seen ever.
  - But, as you will see later, they will serve a useful purpose.
- The circular points are the **points of intersection** of any circle in a plane with l_∞ (the line at infinity).
- Since there can be at most **two intersection points** between a line and a circle, we have exactly **two circular points**.

Circular Points



- Using $\vec{I}$ and $\vec{J}$ to denote the two circular points, we can write:

$$\vec{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, \vec{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

The Principle of Point/Line Duality in Homogeneous Coordinate Formulation of Projective Geometry

- Two of the key results we have derived so far:
  1. The line joining two points x_1 and x_2 is given by $l = x_1 \times x_2$, and
  2. The point of intersection of two lines l_1 and l_2 is $x = l_1 \times l_2$
- These **two keys** should be your first clue as to what we mean by the point/line duality.
  - We can think of the second relation as a **dual** to the first.
- We can take any relation in the homogeneous coordinate formulation of projective geometry and **obtain its dual** by switching points and lines.
  - The dual relation thus obtained is geometrically valid provided you take care in the interpretation of the relation.

Special Entity: The Dual Degenerate Conic C_∞^*

- Recall: a point degenerate conic is defined by two straight lines l and m: $C = lm^T + ml^T$.

- To construct its dual, we must replace each line by a point.
  - You get the dual degenerate conic if you choose the circular points $\vec{I}$ and $\vec{J}$ for this purpose.
  - Denoting the resulting structure by C_∞^* with:
    - the superscript telling us that we are dealing with a dual of a point conic, and
    - the subscript telling us that the anchor points $\vec{I}$ and $\vec{J}$ are at infinity.
  - We get $C_\infty^* = \vec{I}\vec{J}^T + \vec{J}\vec{I}^T$

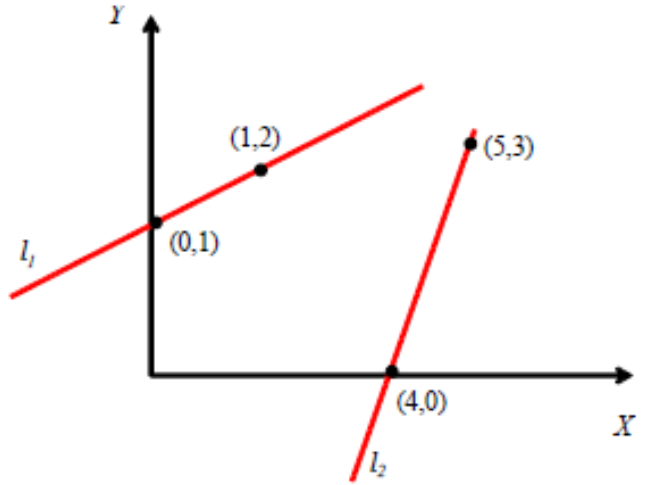
The Dual Degenerate Conic C_∞^*

- For a matrix representation of C_∞^*:
- $C_\infty^* = \vec{I}\vec{J}^T + \vec{J}\vec{I}^T = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \begin{pmatrix} 1 & -i & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} \begin{pmatrix} 1 & i & 0 \end{pmatrix} =$

$$\begin{bmatrix} 1 & -i & 0 \\ i & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & i & 0 \\ -i & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Review Questions

- Derive in just 3 steps the intersection of the two lines shown below:



$$l_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ and } l_2 = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \Longrightarrow \text{Intersection } X = l_1 \times l_2$$

Review Questions

- Why the rank of an outer product matrix always equal to 1?

$$l = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix}, \; m = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} \Longrightarrow lm^T = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix} \begin{pmatrix} m_1 & m_2 & m_3 \end{pmatrix} =$$

$$\begin{bmatrix} l_1 m_1 & l_1 m_2 & l_1 m_3 \\ l_2 m_1 & l_2 m_2 & l_2 m_3 \\ l_3 m_1 & l_3 m_2 & l_3 m_3 \end{bmatrix}$$

Every column is a constant times the first column. *E.g.*, the second col. is m_2/m_1 times the first col. So all cols are linearly dependent, implying a rank of 1.

Review Questions

- Given a conic $x^2 + y^2 - 25 = 0$, where does the tangent to this conic at the perimeter point $(3, 4)$ intersect with the $x - axis$?

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -25 \end{bmatrix} \text{ and } x - axis = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

The tangent line is given $l = CX = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -25 \end{bmatrix} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -25 \end{pmatrix}$,

$\Rightarrow$ The intersect point is $\begin{pmatrix} 3 \\ 4 \\ -25 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$